

# Hussein Ziyard

BSc Applied Data Science Communication (Reading)
General Sir John Kotelawela Defence University

husseinziyard@gmail.com
github.com/Husseinziyard
linkedin.com/in/husseinziyard
husseinziyard.github.io
+94 77 311 3683
Dematagoda, Colombo, Sri Lanka

Data Science undergraduate skilled in Python, R, SQL, and Power BI, with a strong foundation in machine learning and analytics. Proven leader through 2+ years as an Assistant Basketball Coach. Looking for data science opportunities to apply analytical skills to real-world data challenges.

## Experience

**Intern – IT Systems & Innovations** Jul 2026 – Nov 2026
*EFL Logistic Park (LPPL) – 3PL*

- Sorted and resolved WMS system issues by querying and troubleshooting SQL Server databases, including bulk T-SQL updates to fix RF device login errors
- Wrote and debugged SQL queries against the WMS (Körber/HighJump) database to identify and correct data and process issues supporting daily 3PL warehouse operations
- Documented workflows and supported client integration and onboarding discussions for the 3PL system

**Assistant Basketball Coach** Jun 2023 – Present
*Zahira College*

- Assisted training, performance analysis, and team coordination for 15+ players over 2+ years

**Mathematics Teacher (Volunteer)** May 2023 – Sep 2023
*Al Hikma College*

- Delivered lessons and academic support to 20+ students over 5 months, improving comprehension and exam readiness

## Education

**General Sir John Kotelawela Defence University** Jan 2024 – Present
*Bachelor's Degree – Applied Data Science Communication*

**National Institute of Sports Science** Nov 2023 – Dec 2023
*Certificate in Sports Science – Basketball Level 1*

**Zahira College** Feb 2009 – Dec 2022
*Local A-Levels – Combined Mathematics, Physics, ICT*

## Research Papers

**Predicting USD/LKR Exchange Rate: A CRISP-DM ML Approach** 2026
*Presented – ADScAI 2026, University of Moratuwa* *Python, XGBoost, Pandas, NumPy*

- Built 4 ML models (Random Forest, XGBoost, Gradient Boosting, Ensemble) on 16 years of LKR/USD data; XGBoost achieved MAPE of 0.838% and $R^2$ of 0.904 with 30-day recursive forecasting and 95% confidence intervals

**EV Adoption Impact Analysis** 2025
*Research Paper* *Data Analytics*

- Proposed a global analytical model to study environmental and economic impacts of EV adoption across multiple economies

## Academic Projects

**Olympics 2028 Medal Prediction & Dashboard** 2026
*Machine Learning & Data Visualisation* *Python, scikit-learn, Power BI, Excel*

- Ensemble ML model (Random Forest + Gradient Boosting) on 271K+ athlete records (1896–2016) to predict per-country medals for LA 2028; $R^2$ of 0.917 and MAE of 3.46 on held-out 2016 Olympics; engineered lag, rolling, and trend features across 205 countries with 95% confidence intervals; interactive Power BI dashboard with 6 chart types on 3,696 participations and 22K+ medals

**PolicySnap Studio – AI Policy Summarizer & Draft Generator** 2026
*Academic Assignment – Intelligent Systems* *Python, Streamlit, OpenRouter API, pdfplumber*

- Streamlit app that ingests PDF/TXT policy documents, calls an LLM via OpenRouter API to generate structured summaries, and produces scenario-adapted policy drafts (preset + custom scenarios)

**Consumer Behavior, Patient Outcomes & Public Sector Classification** 2024 – 2025
*Group Academic Project – Data Mining* *Python, R, Shiny, Scikit-learn*

- Apriori Association Rule Mining on 500K+ transaction UK retail dataset; Logistic Regression for 30-day hospital readmission prediction on diabetic patient data, evaluated with ROC/AUC; EDA and CRISP-DM clustering on public sector datasets

**Advanced SQL & Cloud Databases: Government & SDG Data** 2025
*Academic Project* *T-SQL, Azure SQL, Power BI, DAX*

End-to-end analytics pipeline on UN SDG datasets (17 Goals, 100+ countries, 2000–2025); Power BI star schema with 5 DAX measures and 5 dashboard pages

**SQL & Power BI: Government & Immigration Analytics** 2024
*SQL Projects – Australian Government Data & Canadian Immigration* *SQL Server, T-SQL, Power BI, DAX*
Relational databases and complex T-SQL queries on Australian government datasets; modelled large-scale Canadian immigration data with Power BI reporting for policy trend analysis

## Events & Leadership

**Co-Chairperson – Data Odyssey 2025** October 2025
*AI and Data Science Club, General Sir John Kotelawela Defence University*
Co-chaired inter-faculty and inter-university data science exhibition; managed logistics for 50+ participants across multiple faculties

**Co-Event Organizer – Data Science Exhibitions & Competitions** Jan 2025 – Feb 2026
*AI and Data Science Club, General Sir John Kotelawela Defence University*
Co-organized 3+ club events including inter-faculty data science workshops and exhibitions

**Chairperson – Inter-House Basketball Tournament** July 2021
*Zahira College Colombo*
Organised and chaired annual tournament, overseeing scheduling, refereeing, and event management

## Technical & Skills

**Programming:** Python, R, SQL

**ML & Algorithms:** Random Forest, XGBoost, Gradient Boosting, Logistic Regression, Apriori, Clustering, Time-Series Forecasting, CRISP-DM

**Libraries & Frameworks:** Pandas, NumPy, Matplotlib, Scikit-learn, Streamlit, Shiny, arules, caret, pROC, ggplot2

**Database & BI:** SQL Server, Azure SQL, T-SQL, Power BI, DAX, Star Schema, ETL

**Tools:** Git, GitHub, VS Code, Jupyter Notebook, Excel

**Soft Skills:** Leadership, Teamwork, Analytical Thinking, Problem-Solving

## Certifications

**Research Paper Presenter – ADScAI 2026** – Applied Data Science & AI Conference, University of Moratuwa, April 2026

**Workshop Participation – ADScAI 2026** – *Event-Driven AI on AWS (Emojot); Shipping Real Features with AI Agents (Cut+Dry)*, April 2026

**Databases and SQL for Data Science with Python** – IBM, February 2026

**Microsoft Power Up Program** – Certified Graduate

**Python for Beginners** – CODL, University of Moratuwa, February 2026

**Certificate in Sports Science – Basketball Level 1** – National Institute of Sports Science, Sri Lanka, March 2024

## Honors & Awards

**Research Paper Presenter – ADScAI 2026** 2026
*Department of Computer Science and Engineering, University of Moratuwa*

**Workshop Participation – ADScAI 2026** 2026
*Department of Computer Science and Engineering, University of Moratuwa*

**Champions – Data Odyssey Inter Faculty Exhibition Competition 2025** 2025
*AI and Data Science Club, General Sir John Kotelawela Defence University*

**KDU Merit Award for Basketball – Colours Night 2025** 2025
*General Sir John Kotelawela Defence University*

**Champions – Green Basketball Championship 2025** 2025
*NSBM Basketball Club, NSBM Green University*

**Participant – Data Odyssey Inter-University Exhibition Competition 2024** 2024
*AI and Data Science Club, General Sir John Kotelawela Defence University*

**Best Basketball Player – Annual Prize Day** 2022
*Zahira College*

**University Entrance Prize – Annual Prize Day** 2022
*Zahira College*

## References

**Dr. Charith Silva**
Senior Lecturer
General Sir John Kotelawela Defence University
+44 7920 206338
charithsilva@kdu.ac.lk

**Mr. Fayaz Mooseen**
Chief Executive Officer
Textus Technology (Pvt) Ltd.
+94 77 229 2281
fayazlk@gmail.com